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Via-filling method of through-glass via substrate

Abstract

A via-filling method of a TGV substrate includes steps: filling a plurality of metal balls into a plurality of vias of the TGV substrate; using a heating process to melt the plurality of metal balls to form a liquid-state metal; and cooling down the liquid-state metal to form a solid-state metal inside the plurality of vias. Because the method needn't use solvents or fluxes, the solid-state metal inside the plurality of vias have better electric conductivity.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5240671	12/1992	Carey	264/633	C04B 41/4578
7772123	12/2009	Birner	257/E21.585	H01L 21/76877
9305866	12/2015	Lu	N/A	H01L 23/49894
9379043	12/2015	Wang	N/A	H01L 21/76898
10756009	12/2019	Hoffmeyer	N/A	H05K 3/308
2011/0266671	12/2010	Youm	257/738	H01L 23/49827
2021/0125897	12/2020	Valavala	N/A	H10N 19/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
9404248	12/1994	BR	B23K 1/0008
19608683	12/1996	DE	B23K 35/0222
1387401	12/2003	EP	H01L 21/44
3007199	12/2013	FR	H01L 23/49827
20130030988	12/2012	KR	H05K 3/421
WO-9853494	12/1997	WO	H01L 21/4814
WO-2011136403	12/2010	WO	H01L 23/49827

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Background/Summary

(1) This application claims priority of application Ser. No. 11/133,007 filed in Taiwan on 31 Aug. 2022 under 35 U.S.C. § 119; the entire contents of all of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a Through-Glass Via (TGV) technology, particularly to a via-

filling method of a TGV substrate.

Description of the Prior Art

(3) 3D package technology has been extensively developed in recent years. In a 3D package structure, a TGV substrate is usually used as an interposer to electrically connect an upper chip above the TGV substrate with a lower chip below the TGV substrate. The interposer may also be called the intervening layer or the intermediate layer. The upper chip is electrically connected with the lower chip through a solid-state metal inside a plurality of vias of the TGV substrate. The current via-filling methods of the TGV substrate include using a printing method to fill a metal paste into a plurality of vias of the TGV substrate, and then sintering the metal paste to form a solid-state metal inside the plurality of vias. In the traditional via-filling technology, a solvent or a flux is added to the metal paste to keep the flowability of the metal paste and make the metal paste filled easily. However, the solvent and flux are less likely to vaporize. Thus, pores may appear inside the sintered solid-state metal and lead to poor electric conductivity.

SUMMARY OF THE INVENTION

(4) The present invention provides a via-filling method of a TGV substrate for enhancing the electric conductivity.

(5) In one embodiment, a via-filling method of a TGV substrate of the present invention comprises Step A: coating a release agent on the upper surface of a fixture; Step B: placing the TGV substrate on the upper surface of the fixture, wherein the TGV substrate has a plurality of vias; Step C: filling a plurality of metal balls into the plurality of vias; Step D: melting the plurality of metal balls to form a liquid-state metal; Step E: cooling the liquid-state metal inside the plurality of vias to generate a solid-state metal; Step F: repeating Steps C-E until the plurality of vias is fully filled by the solid-state metal; and Step G: cleaning the surface of the TGV substrate, and removing the TGV substrate from the fixture.

(6) In comparison with the conventional via-filling technology using a metal paste, the present invention needn't use a solvent or a flux. Therefore, the solid-state metal inside the vias is free of pores and has better electric conductivity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top view schematically showing a TGV substrate where vias have not been filled yet according to one embodiment of the present invention.

(2) FIG. 2 includes a top view and a sectional view schematically showing a fixture according to one embodiment of the present invention.

(3) FIG. 3 is a diagram schematically showing that a TGV substrate is placed on a fixture according to one embodiment of the present invention.

(4) FIG. 4 is a diagram schematically showing that metal balls are poured onto a TGV substrate according to one embodiment of the present invention.

(5) FIG. 5 includes a top view and a sectional view schematically showing that metal balls are filled into the vias of a TGV substrate according to one embodiment of the present invention.

(6) FIG. 6 includes a top view and a sectional view schematically showing that metal balls are melted and then cooled down to form a solid-state metal according to one embodiment of the present invention.

(7) FIG. 7 includes a top view and a sectional view schematically showing that vias are fully filled by a solid-state metal according to one embodiment of the present invention.

(8) FIG. 8 includes a top view and a sectional view schematically showing that a TGV substrate whose vias are fully filled according to one embodiment of the present invention.

(9) FIG. 9 is a flowchart of a via-filling method of a TGV substrate according to one embodiment

of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(10) FIGS. 1-8 are used to demonstrate the via-filling method of a TGV substrate according to one embodiment of the present invention. Firstly, acquire a TGV substrate **10**, which has a plurality of vias **12** interconnecting the upper surface and the lower surface of the TGV substrate **10**, as shown in FIG. 1. Next, coat a heat-resistant release agent **22** on the upper surface of a fixture **20**, wherein the release agent **22** makes the TGV substrate **10** removed from the fixture **20** more easily, as shown in FIG. 2. In FIG. 2, the upper drawing is a top view of the fixture **20**, and the lower drawing is a sectional view of the fixture **20**. After the release agent **22** has been coated, place the TGV substrate **10** on the upper surface of the fixture **20**, as shown in FIG. 3.

(11) After the TGV substrate **10** has been placed on the fixture **20**, fill a plurality of metal balls **30** into the plurality of vias **12** of the TGV substrate **10**, as shown in FIG. 4 and FIG. 5, wherein the plurality of metal balls **30** is poured onto the TGV substrate **10** firstly (as shown in FIG. 4), and then the fixture **20** is slightly vibrated or swung to make the metal balls **30** roll and fall into the vias **12** (as shown in FIG. 5). In FIG. 5, the upper drawing is a top view of the TGV substrate **10** and the fixture **20**, and the lower drawing is a sectional view taken along Line AA' in the top view. There are many methods to fill the metal balls **30** into the vias **12**, and the method shown in FIG. 4 and FIG. 5 is only one of them. In the embodiment shown in FIG. 4 and FIG. 5, the height and radius of the via **12** are respectively H and $R1$, and the radius of each metal ball **30** is $R2$, wherein the height H of the via **12** is exactly the thickness of the TGV substrate **10** and the radius $R1$ of the via **12** is greater than or equal to the radius $R2$ of the metal ball **30**. The volume of each via **12** is $H \times \pi \times R1^2$, and the volume of each metal ball **30** is $(4/3) \times \pi \times R2^3$, wherein π is the circular ratio. In the present invention, the remainder of the volume of the via **12** divided by the volume of the metal ball **30** is expected to approach zero as much as possible. In other words, it is the optimized condition: the volume of each via **12** is an integer multiple of the volume of each metal ball **30**. In the case that the height H and the radius $R1$ of the via **12** have been known, an appropriate radius $R2$ of the metal ball **30** can be designed. In one embodiment, the height H and the diameter $D1$ of the via **12** ($D1=2 \times R1$) are respectively $200\ \mu\text{m}$ and $125\ \mu\text{m}$ and the diameter $D2$ of the metal ball **30** ($D2=2 \times R2$) is $105\ \mu\text{m}$.

(12) After a plurality of metal balls **30** has been filled into the plurality of vias **12** of the TGV substrate **10**, perform a heating process on the plurality of metal balls **30** inside the vias **12** to melt the metal balls **30** to form a liquid-state metal. After the plurality of metal balls **30** has been completely liquidized, perform a cooling process to cool down the liquid-state metal inside the plurality of vias **12** and generate a solid-state metal **32**, as shown in FIG. 6.

(13) In one embodiment, the heating process includes a reflow process. The reflow process includes a first heating stage, an isothermal stage, and a second heating stage. The reflow process heats the entire TGV substrate **10**. In the first heating stage, the temperature of the TGV substrate **10** is raised from a first temperature to a second temperature at a heating rate of $1-2^\circ\text{C. per second}$. After the TGV substrate **10** has reached the second temperature, the process terminates the first heating stage and enters the isothermal stage. In the isothermal stage, the process maintains the TGV substrate **10** at the second temperature. After the isothermal stage is terminated, the process enters the second heating stage. In the second heating stage, the temperature of the TGV substrate **10** is raised from the second temperature to a third temperature at a heating rate of $1-2^\circ\text{C. per second}$. Once the TGV substrate **10** reaches the third temperature, the reflow process is terminated. In one embodiment, the first temperature may be but is not limited to be 25°C. , the second temperature may be but is not limited to be 180°C. , and the third temperature may be but is not limited to be 230°C.

(14) In one embodiment, the heating process includes a laser heating process. The laser heating process uses a laser beam to irradiate the plurality of metal balls **30** and melt the plurality of metal balls **30**. In general, a laser beam can only irradiate a local region of the TGV substrate **10**.

Therefore, the laser heating process lets the laser beam irradiate a plurality of regions of the TGV substrate **10** in sequence to melt the metal balls **30** inside all the vias **12**. In one embodiment, the laser heating process uses a laser beam with a power of 6-8 W, and each region of the TGV substrate **10** is irradiated by the laser beam for 5-9 seconds.

(15) In comparison with the conventional sintering process, the reflow process or the laser heating process, which is used by the present invention, can reduce the heating time. Thus, the present invention can decrease the probability of damaging the TGV substrate **10** during heating.

(16) If the solid-state metal **32** inside the vias **12** of the TGV substrate **10** have not fully filled yet (as shown in FIG. 6), the steps shown in FIGS. 4-6 are repeated until all the vias **12** of the TGV substrate **10** have been fully filled by the solid-state metal, as shown in FIG. 7. After all the vias **12** of the TGV substrate **10** have been fully filled by the solid-state metal, a cleaning agent is used to clean the surface of the TGV substrate **10**, wherein the cleaning agent may be but is not limited to alcohol. After cleaning, the TGV substrate **10** is removed from the fixture **20**, and the via-filling process of the TGV substrate **10** is completed, as shown in FIG. 8. In comparison with the conventional via-filling technology that fills vias with metal paste, the present invention is characterized in filling the vias **12** with solid-state metal balls and then melting the metal balls. Therefore, the present invention needn't use a solvent or a flux, exempts the solid-state metal **32** from pores, and enhances the electric conductivity of the solid-metal **32**.

(17) From the above description, it is learned: the via-filling method of a TGV substrate of the present invention can be expressed by FIG. 9 and comprises: Step S10: coating a release agent on the upper surface of a fixture; Step S11: placing a TGV substrate on the upper surface of the fixture; Step S12: filling a plurality of metal balls into a plurality of vias of the TGV substrate; Step S13: melting the plurality of metal balls to form a liquid-state metal; Step S14: cooling the liquid-state metal inside the plurality of vias to generate a solid-state metal; Step S15: determining whether the plurality of vias are fully filled by the solid-state metal; if no, repeating Steps S12-S14; if yes, proceeding to Step S16; and Step S16: cleaning the surface of the TGV substrate, and removing the TGV substrate from the fixture.

(18) The embodiments described above are only to exemplify the present invention but not to limit the scope of the present invention. Through making some modification or variation according to the technical contents disclosed in the specification and claims, any person having ordinary knowledge of the art should be able to generate equivalent embodiments without departing from the present invention. Further, the equivalent embodiments are to be included by the scope of the present invention.

Claims

1. A via-filling method of a Through-Glass Via (TGV) substrate, comprising: Step A: coating a release agent on an upper surface of a fixture; Step B: placing the TGV substrate on the upper surface of the fixture, wherein the TGV substrate has a plurality of vias; Step C: filling a plurality of metal balls into the plurality of vias; Step D: melting heating the plurality of metal balls to melt the plurality of metal balls to form a liquid-state metal; Step E: cooling the liquid-state metal inside the plurality of vias to generate a solid-state metal; Step F: repeating Steps C-E until the plurality of vias is fully filled by the solid-state metal; and Step G: cleaning a surface of the TGV substrate, and removing the TGV substrate from the fixture; wherein in Step D the heating includes a reflow process comprises a first heating stage, a second heating stage to raise a temperature of the TGV substrate and an isothermal stage, arranged between the first heating stage and the second heating stage to maintain the temperature of the TGV substrate.

2. The via-filling method according to claim 1, wherein the reflow process further comprises: a first heating stage: raising the temperature of the TGV substrate from a first temperature to a second temperature at a rate of 1 to 2 degrees per second in the first temperature stage; and a second

- heating stage: raising the temperature of the TGV substrate from the second temperature to a third temperature at a rate of 1 to 2 degrees per second in the second temperature stage.
3. The via-filling method according to claim 2, wherein the reflow process further comprises: an isothermal stage, arranged between the first heating stage and the second heating stage, wherein in the isothermal stage, the temperature of the TGV substrate is maintained at the second temperature.
 4. The via-filling method according to claim 2, wherein the first temperature is 25° C., the second temperature is 180° C., and the third temperature is 230° C.
 5. The via-filling method according to claim 1, wherein in Step D the heating further comprises a step: using a laser heating process to melt the plurality of metal balls.
 6. The via-filling method according to claim 5, wherein the laser heating process uses a laser beam with a power of 6-8 W to irradiate a plurality regions of the TGV substrate in sequence to melt the plurality of metal balls, wherein each of the plurality of regions is irradiated by the laser beam for 5-9 seconds.
 7. The via-filling method according to claim 1, wherein a volume of each of the plurality of vias is an integer multiple of a volume of each of the plurality of metal balls.
 8. The via-filling method according to claim 1, wherein a height of each of the plurality of vias and a diameter of each of the plurality of vias are respectively 200 μm and 125 μm, and a diameter of each of the plurality of metal balls is 105 μm.
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